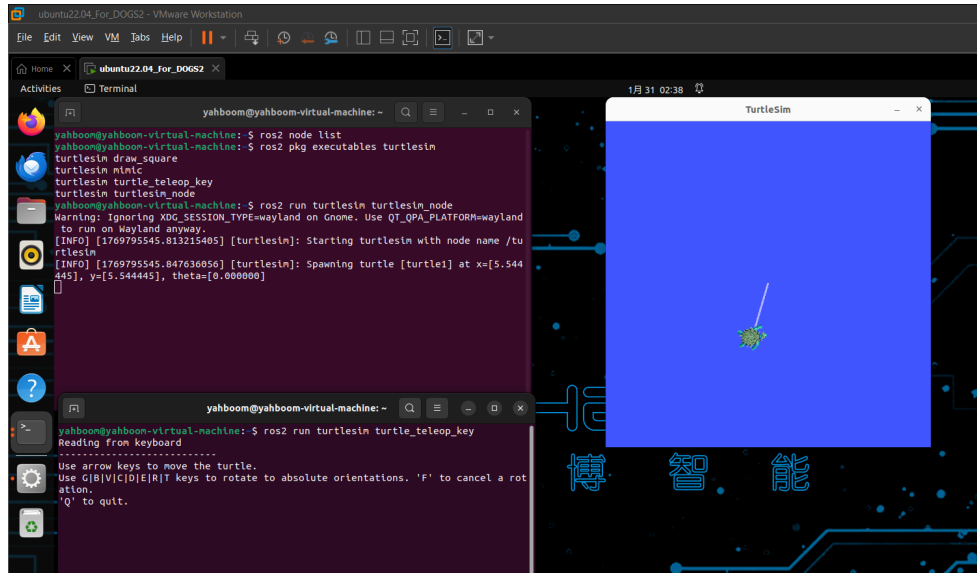
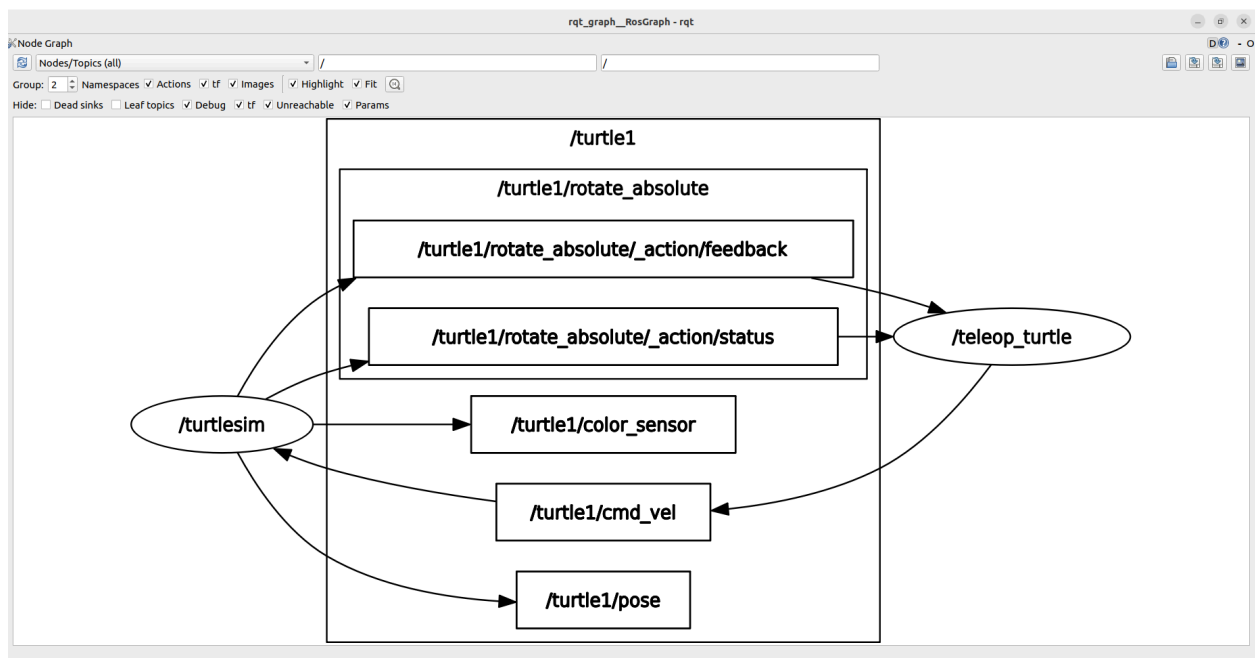


Task 2.1 Starting Nodes:



Task 2 Topics Step 7:



Task 2 Topics Step 21:

```

yahboom@yahboom-virtual-machine:~$ ros2 topic info /turtle1/pose
Type: turtlesim/msg/Pose
Publisher count: 1
Subscription count: 0
yahboom@yahboom-virtual-machine:~$ ros2 topic echo /turtle1/pose
x: 5.544444561004639
y: 5.544444561004639
theta: 0.0
linear_velocity: 0.0
angular_velocity: 0.0
---
x: 5.544444561004639
y: 5.544444561004639
theta: 0.0
linear_velocity: 0.0
angular_velocity: 0.0
---
x: 5.544444561004639
y: 5.544444561004639
theta: 0.0
linear_velocity: 0.0
angular_velocity: 0.0
---
x: 5.544444561004639
y: 5.544444561004639
theta: 0.0

```

```

yahboom@yahboom-virtual-machine:~$ ros2 interface show turtlesim/msg/Pose
float32 x
float32 y
float32 theta

float32 linear_velocity
float32 angular_velocity

```

Task 2 Topics Step 25:


```

ros2: error: unrecognized arguments: type
yahboom@yahboom-virtual-machine:~$ ros2 action send_goal /turtle1/rotate_absolute turtlesim/action/RotateAbsolute "{theta: 1.57}" --feedback
Waiting for an action server to become available...
Sending goal:
  theta: 1.57
Feedback:
  remaining: -0.44599997997283936
Goal accepted with ID: 9441981b200b4ff89c6fd45f44f39d17
Feedback:
  remaining: -0.4299999475479126
Feedback:
  remaining: -0.41399991512298584
Feedback:
  remaining: -0.39800000190734863
Feedback:
  remaining: -0.3819999694824219
Feedback:
  remaining: -0.3659999370574951
Feedback:
  remaining: -0.34999990463256836
Feedback:
  remaining: -0.3339999141693115
Feedback:
  remaining: -0.3179999589920044
Feedback:
  remaining: -0.30199992656707764
Feedback:
  remaining: -0.2859998941421509
Feedback:
  remaining: -0.26999998092651367
Feedback:
  remaining: -0.2539999485015869
Feedback:

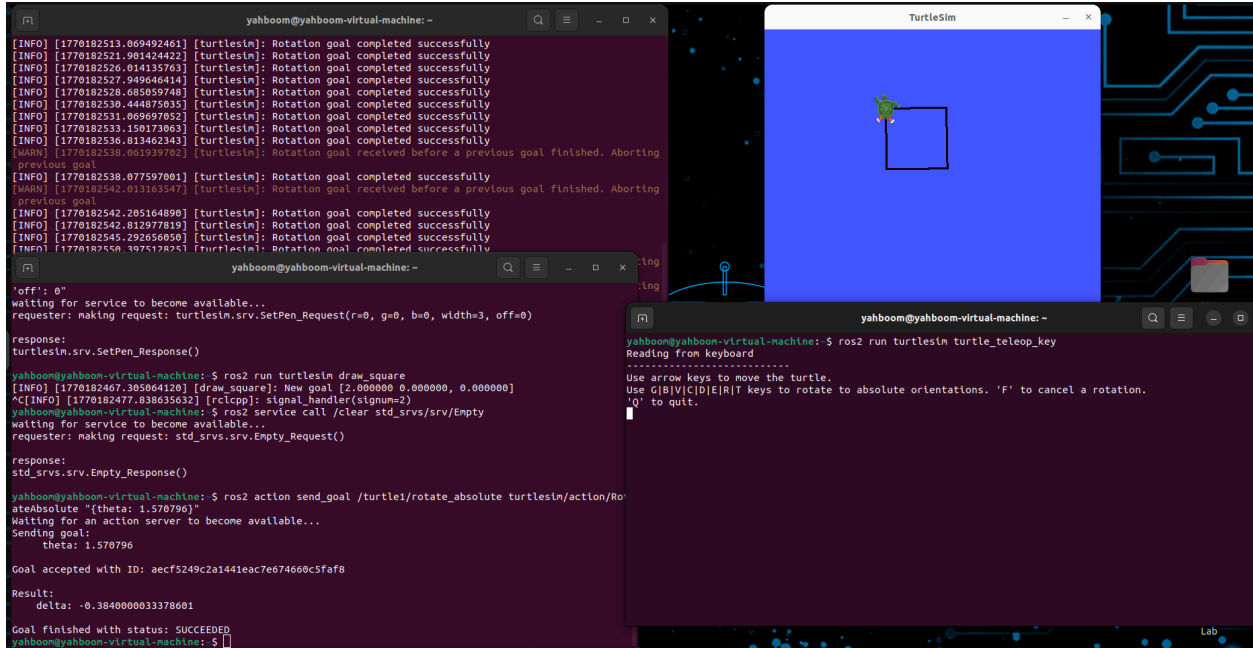
```

```

Feedback:
  remaining: -0.046000003814697266
Feedback:
  remaining: -0.029999971389770508
Feedback:
  remaining: -0.01399993896484375
Result:
  delta: 0.4320000112056732
Goal finished with status: SUCCEEDED
yahboom@yahboom-virtual-machine:~$

```

Task 3:



For task 3, I used `ros2 service call /clear_std_srvs/srv/Empty` to clear any previously made patterns. I then set the pen color to black with a width of 3 using:

```
ros2 service call /turtle1/set_pen turtlesim/srv/SetPen "'r': 0
'g': 0
'b': 0
'width': 10
'off': 0"
```

I then used the absolute rotations, keyboard keys, and arrow keys to steer the turtle in the right directions to make a square.